

Assignment 3

1. 真值表:

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

注: 尽量按照递增
的顺序写真值表,

000 $\rightarrow$ 001 $\rightarrow$

010 $\rightarrow \dots \rightarrow$ 111

布尔表达式: $Y = \overline{AB} \cdot \bar{C}$

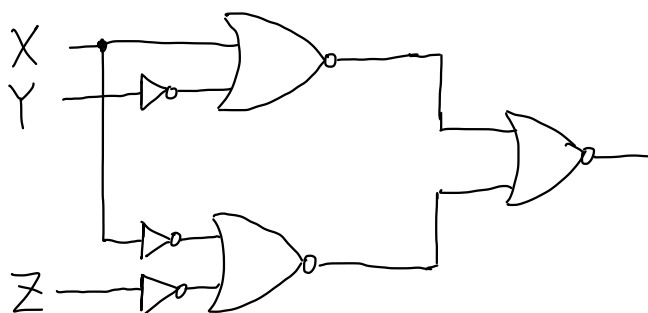
$$= (\bar{A} + \bar{B}) \cdot \bar{C}$$

注: $\overline{AB}$ 是指对 A 和 B
与之后的结果取反,
不是 $\bar{A} \cdot \bar{B}$,

$$\overline{AB} \neq \bar{A} \bar{B}$$

2.

a)



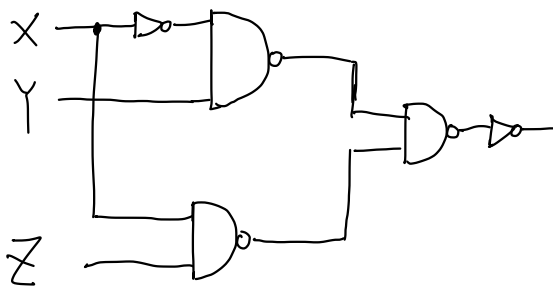
b) 对布尔表达式变形。(利用摩根律)

$$\overline{[(X + \bar{Y}) + (\bar{X} + \bar{Z})]}$$

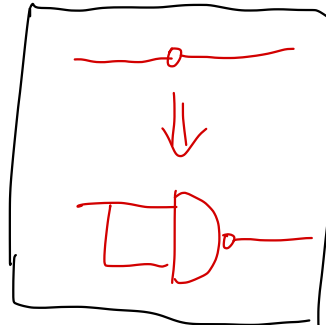
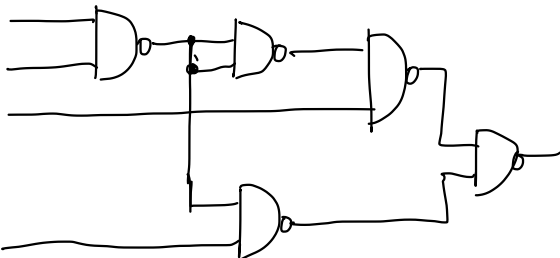
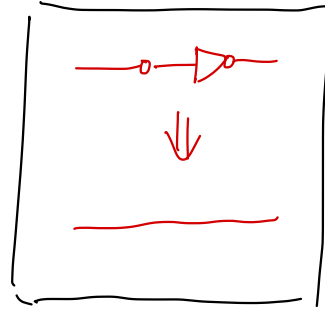
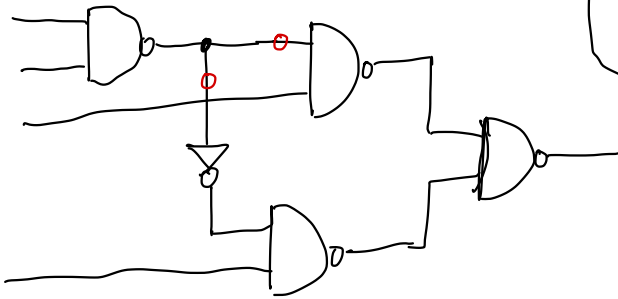
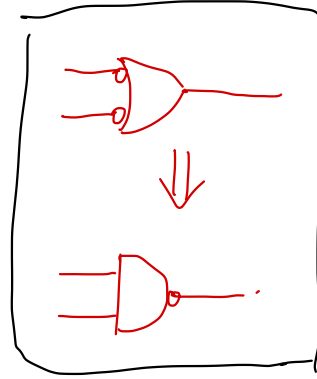
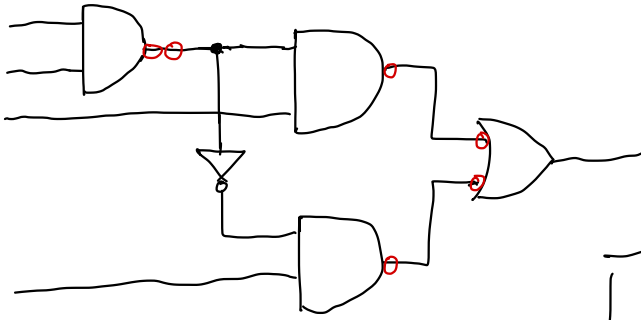
$$= \overline{[(\bar{X} \cdot Y) + (XZ)]}$$

$$= \overline{(\bar{X}Y)} \cdot \overline{(XZ)}$$

$$= \overline{\bar{X}Y} \cdot \overline{XZ}$$



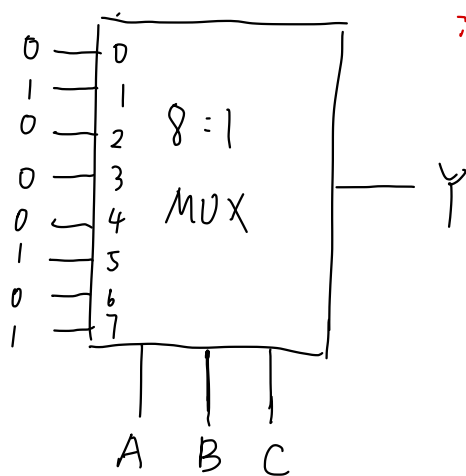
3.



4. 先写真值表, 再画电路图实现

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

(a)



注: 编码一定要写出来、

编码规则:

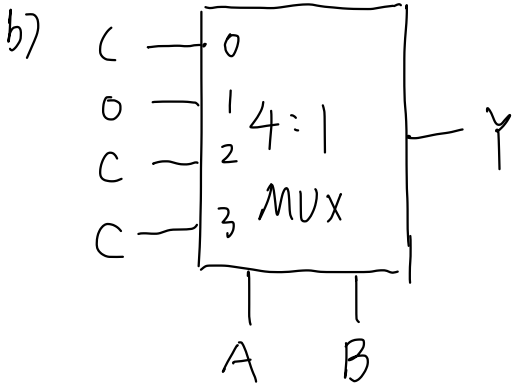
ABC 取值为 000 时,

编码为其对应的十进制

即可.

$ABC = 001 \Rightarrow 1$

$ABC = 111 \Rightarrow 7$



注：从真值表^V选取 $n-1$ 个变量，
中

看剩下的一个变量与输入
之间的关系：

c) 最简单的方法就是根据前面的 4 选 1 改造一下，
先用 A 做选择，再用 B 做选择

